Does Splitting Returns with Wavelets Help GARCH Forecast Volatility? A Re-implementation and Extension on Six Markets

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Abstract

Forecasting volatility is important for risk management, but daily returns are noisy, fat tailed and their variance keeps changing. Our assigned paper, Ismail et al. (2016), first transforms the returns with a Haar-type MODWT, fits ARMA and GARCH(1,1) to the transformed series, and compares this MODWT-GARCH(1,1) with a plain GARCH(1,1) on four African stock indices. They report that MODWT-GARCH gives the better volatility forecasts. In this project we wrote the whole method again from scratch in base R and tested it in a strict rolling, one-step-ahead setup with no look-ahead. We use three of the paper's African markets (South Africa, Tunisia, Nigeria) and three stocks (Infosys, IBM, Google). Training is 2004–2009, validation is 2010–2011 and the test is 2012–2014. The stocks also get a later check on 2015–2025. All three pipelines (raw returns, paired Haar DWT and Haar MODWT) use an ARMA mean picked by AIC and a GARCH(1,1) variance, and they all forecast the same target, the next squared deviation from the past 20-day mean.

We then added things that the paper did not do: Student-t and GED innovations (because every series fails the Jarque–Bera normality test, the kurtosis goes from 5.2 to 522), a structural MODWT order rule, 24 wavelet component models on the target itself, three equal-weight ensembles, a small neural network (MLP) with three seeds, and a sensitivity study on window, refit cadence, information criterion and order cap.

Our main finding is that the wavelet step helps much less than the paper suggests. On the 2012–2014 test, MODWT-GARCH is within 3% MSE of plain GARCH on all six series. A paired Haar DWT version is better on Nigeria (9.3% lower MSE), Tunisia (7.9%, but with a wide uncertainty interval) and Infosys (3.0%), and about the same on the other three. Fat-tailed innovations give small but steady gains: Student-t or GED has the lowest MSE in 23 of 27 held-out groups, and the gain is largest on Tunisia, the fattest-tailed series. Ensembles are mixed, and the MLP is worse than GARCH everywhere. So our honest conclusion is that a well-tuned GARCH(1,1) is very hard to beat, and wavelets are a useful option for some markets but not a general improvement.

1. Introduction

Volatility tells us how much a price moves around. Banks and fund managers use volatility forecasts for Value-at-Risk, for portfolio weights and for option pricing, so a better forecast has a direct money value. The problem is that daily returns are mostly noise. The mean is close to zero, big moves come in clusters (a calm 2005 and a wild 2008), and the distribution has much fatter tails than a normal one.

The classic tool is GARCH (Engle, 1982; Bollerslev, 1986). It models today's variance from yesterday's shock and yesterday's variance, and it captures clustering quite well. Ismail et al. (2016) argue that a return series mixes moves at many time scales, from day-to-day noise to slow trends, and that a wavelet filter can denoise the returns before GARCH is fitted. They test this on four African stock indices (Kenya, Nigeria, South Africa, Tunisia, 2000–2014) and find that MODWT-GARCH(1,1) has lower AIC on three of the four markets and smaller forecast errors than GARCH(1,1) on all four. Rubio et al. (2023) report a similar gain from wavelet + ARIMA–GARCH hybrids for the volatility of Tesla stock.

Our goal was to check this claim carefully. Our plan was:

1. Re-build the three pipelines of the paper (raw, DWT, MODWT with ARMA–GARCH) ourselves, in base R, with no wavelet or GARCH packages, so we understand every step.
2. Test them in a fair way: one-step-ahead rolling fore-

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casts, every model sees only the past, all model choices are frozen on a validation period before the test, and all models are scored on exactly the same dates.

3. Use three of the paper's four African markets plus three large stocks from India and the USA, to see if the result travels. (Our Kenya NSE-20 data only starts in 2010, which is too short for our training period, so Kenya is left out.)

4. Add our own extensions: fat-tailed innovations, component models, ensembles, an MLP and a sensitivity study.

The rest of the report is like this. Section 2 reviews the background. Section 3 explains the method of the paper as we understood and implemented it. Section 4 describes our extensions. Section 5 gives the experiments and results and Section 6 concludes.

2. Background and Related Work

ARMA. An ARMA(p, q) model writes the return as a linear function of its own past values and past shocks, $\Phi(B)r_t = c + \Theta(B)\varepsilon_t$ (Box et al., 2015). It is good for the mean, but it assumes the shock variance is constant, so it can not describe volatility clustering.

GARCH. Engle (1982) let the variance depend on past squared shocks (ARCH) and Bollerslev (1986) added the past variance itself. For GARCH(1,1),

$$\varepsilon_t = \sqrt{h_t}\, z_t, \qquad h_t = \omega + \alpha\varepsilon_{t-1}^2 + \beta h_{t-1}, \qquad (1)$$

with $\omega > 0$, $\alpha, \beta \geq 0$ and $\alpha + \beta < 1$ for a finite long-run variance. Hansen & Lunde (2005) compared hundreds of volatility models and found that it is surprisingly hard to beat GARCH(1,1), which is something we also saw. The standard choice is Gaussian z_t, but real returns have fat tails, so Bollerslev (1987) used a Student-t and Nelson (1991) used the generalized error distribution (GED).

Wavelets. A wavelet transform splits a signal into a smooth approximation A (low frequency) and details D (high frequency) by passing it through a low-pass and a high-pass filter (Mallat, 1989). The discrete wavelet transform (DWT) keeps every second value after filtering, so each level halves the length. The maximal overlap DWT (MODWT) does not downsample. It keeps one coefficient per time point, works for any length and does not depend on where the series starts (Percival & Walden, 2000). With the Haar filter, the simplest one, the first-level coefficients are just scaled sums and differences of neighbouring values.

Wavelet + GARCH hybrids. Ismail et al. (2016) compute a first-level Haar-type wavelet series from the returns, $w = (r_{2k} - r_{2k-1})/\sqrt{2}$, choose an ARMA order for it by AIC (they also report serial-correlation and ARCH-LM tests for every candidate), fit GARCH(1,1), and compare one-step variance forecasts with plain ARMA–GARCH(1,1) using MSE against a volatility proxy, the squared deviation of the return from a moving-average expected return. They also report moments, Ljung–Box $Q(24)$ and Jarque–Bera tests of the returns. Rubio et al. (2023) decompose Tesla volatility with the DWT into approximation and detail parts, model them with ARIMA and GARCH in two hybrid ways, and check normality (Jarque–Bera, skewness, kurtosis), stationarity (ADF, KPSS) and independence (BDS) first. Both papers report that the hybrids do better. One thing we noticed is that the papers do not say exactly how the forecast of a wavelet coefficient is turned back into a forecast of the return variance. This step turned out to be the most important design choice in our work.

Measuring accuracy. True volatility is never observed, so we need a proxy. Patton (2011) shows that MSE on a squared-return type proxy still ranks models correctly in expectation, even though the proxy is noisy. We use MSE (and RMSE) on such a proxy for this reason.

3. Approach: the Paper's Method as We Implemented It

3.1. Target and rolling protocol

Let P_t be the closing price and $r_t = \ln P_t - \ln P_{t-1}$ the daily log return. All models forecast the same target,

$$y_{t+1} = \left(r_{t+1} - \bar{r}_t^{(20)}\right)^2, \quad \bar{r}_t^{(20)} = \tfrac{1}{20}\textstyle\sum_{i=0}^{19} r_{t-i}, \qquad (2)$$

which is the squared deviation of tomorrow's return from the mean of the last 20 returns. This follows the volatility proxy of Ismail et al. (2016), the squared deviation from a moving-average expected return; the 20-day window is our choice. The mean $\bar{r}_t^{(20)}$ is known at time t, so there is no leakage. Using one target for everyone is the only way to compare raw, DWT and MODWT models fairly.

If a model gives a conditional mean μ_{t+1} and variance h_{t+1} for the next return, the forecast of the target is

$$\hat{y}_{t+1} = h_{t+1} + \left(\mu_{t+1} - \bar{r}_t^{(20)}\right)^2, \qquad (3)$$

because $E[(r - m)^2] = \mathrm{Var}(r) + (Er - m)^2$.

Forecasts are made one step ahead, every trading day. Each model is fitted on a rolling window of the last 1,024 observations and is re-estimated every 20 days. Between refits the parameters stay fixed but the filter still updates with every new observation. A forecast for day $t + 1$ only uses data up to day t. This is checked by a leakage test in our code: changing future values does not change any past forecast.

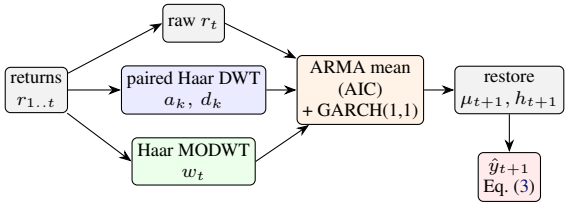


Figure 1. The three core pipelines. Each input branch gets its own ARMA–GARCH, and the forecast is mapped back to the common target.

3.2. Mean and variance models

For every input series we pick the ARMA(p, q) order with $p, q \leq 3$ by AIC $= -2\ell + 2k$ (Akaike, 1974) on the last 1,024 training returns. The ARMA is estimated by Gaussian maximum likelihood. Then a GARCH(1,1) (Eq. 1) is fitted on the ARMA residuals. So it is a two-step estimate and not a joint ARMA–GARCH likelihood; we chose this because it is simpler and more stable. The orders picked for the raw returns at the end of training were (3,1) for South Africa, (2,0) Tunisia, (0,3) Nigeria, (0,3) Infosys, (2,2) IBM and (3,2) Google. The GARCH persistence $\alpha + \beta$ was between 0.93 and 0.997, so volatility is very persistent in all six series.

3.3. The three pipelines

Figure 1 shows the three pipelines. The paper's wavelet formula $w = (r_{2k} - r_{2k-1})/\sqrt{2}$ uses non-overlapping pairs of returns, which is the first-level Haar DWT detail, while the name MODWT normally means the overlapping (undecimated) version. Since the paper is not fully clear here, we implemented both: the paired Haar DWT exactly as in their formula, and the overlapping Haar MODWT. The rolling protocol, the restoration step and the common target are our own design choices, needed to turn the method into a fair out-of-sample forecast.

Raw GARCH. ARMA–GARCH directly on r_t. This is the benchmark.

MODWT-GARCH. The first-level Haar MODWT detail is $w_t = (r_t - r_{t-1})/2$. We model w_t with ARMA–GARCH. Since $r_{t+1} = r_t + 2w_{t+1}$ and r_t is already known, the return moments are $\mu_r = r_t + 2\mu_w$ and $h_r = 4h_w$.

Paired DWT-GARCH. The DWT works on non-overlapping pairs (u_k, v_k) of returns, giving an approximation $a_k = (u_k + v_k)/\sqrt{2}$ and a detail $d_k = (v_k - u_k)/\sqrt{2}$. These two series are half as long as the returns. We fit one ARMA–GARCH to each, which gives one-step moments (μ_a, h_a) and (μ_d, h_d) for the next pair (r_t, r_{t+1}). The tricky part is that the first return r_t of that pair is already observed when we forecast r_{t+1}. So we should condition on it. Because Haar is orthonormal we treat a and d as conditionally uncorrelated and jointly normal. Then $s = a - d = \sqrt{2}\, r_t$ is known and

$$E[d \mid s] = \mu_d - \frac{h_d}{h_a + h_d}\big(s - (\mu_a - \mu_d)\big), \tag{4}$$

$$\mu_r = r_t + \sqrt{2}\, E[d \mid s], \qquad h_r = \frac{2h_a h_d}{h_a + h_d}. \tag{5}$$

A good sanity check: if the returns are i.i.d. with mean m and variance σ^2, then $h_a = h_d = \sigma^2$, $\mu_a = \sqrt{2}m$, $\mu_d = 0$, and the formulas give back $\mu_r = m$ and $h_r = \sigma^2$, the same as the raw model. Our unit tests check this on simulated data. If we ignore that r_t is known and just use $h_r = 2h_d$, the variance forecast is too large by a big factor, so this conditioning step really matters.

3.4. Evaluation

The main score is the MSE (and RMSE) of $\hat{y}$ against y on the test days. We report the gain of a model over raw GARCH as $100\,(1 - \mathrm{MSE}_{\text{model}}/\mathrm{MSE}_{\text{raw}})$, so positive means better. Some fits fail on some days (the optimiser does not converge). Because of this, every comparison is done on matched dates only: the dates on which all models in that table have a forecast. This is why the same model can have a slightly different number in two tables; each table says which dates it uses. As a rough guide for uncertainty we also compute paired moving-block bootstrap intervals (Künsch, 1989), but we use them only as a description, not as a formal test.

4. Our Extensions

The paper only assumes Gaussian distribution on GARCH on African indices. We added the following extensions. The paper chose 2010-2011 validation period and settings were kept the same for test set.

E1: Student-t distribution and GED. Our tests (Table 1) show that all six datasets reject normality test, so using a Gaussian z_t in Eq. (1) is not accurate. We again fit each of the three pipelines using a standardized Student-t (shape ν) distribution and with a GED (shape κ), keeping the same ARMA mean, window and procedure. We calculated likelihoods for both process using `optim`.

E2: structural MODWT order. Multiplying a raw ARMA(p, q) by the Haar filter $(1 - B)/2$ adds one MA lag, so the MODWT detail should follow ARMA$(p, q + 1)$. We tried this rule (with Gaussian and Student-t) as an alternative to choosing the MODWT order by AIC. The validation period picks between them per series.

E3: wavelet component models of the target. Instead of transforming returns, we also decompose the target y itself into A_J and $D_J, \ldots, D_1$, forecast each part and sum them.The approximation is put into ARMA and the detail

part is put into GARCH.

E4: Ensemble learning. We built three ensembles, without fitting any weights on the test: *base* (the six raw/DWT/MODWT ARMA and GARCH forecasts), *combined* (base plus the direct-target ARMA and four selected component models) and *MODWT* (two MODWT return models and two MODWT component models).

E5: Neural network. We also implement a hidden-layer MLP with 7 inputs (the last five targets and the past 5-day and 20-day mean), 3 or 5 sigmoid units, weight decay 0.001 or 0.01, and a linear output.We test over three seeds (101, 202, 303) and use validation MSE to determine size and decay, and we report all three seeds on the test.

E6: Change in settings. For the three pipelines we change one setting at a time: window 768 or 1,536 (instead of 1,024), refit every 40 or 60 days (instead of 20), use BIC instead of AIC.

E7: More markets and a later period. Besides three of the paper's African indices we add Infosys (India), IBM and Google (USA), and for the stocks we also run the models on 2015–2025.

5. Experiments

5.1. Data and setup

The daily closing prices were obtained from the following public data sources : FTSE/JSE All Share (South Africa), TUNINDEX (Tunisia) and the NGX All Share Index (Nigeria), and Yahoo Finance adjusted closes for Infosys, IBM and Google. We kept all trading dates and did not interpolate or fill the data. We kept a record of the inconsistencies in the data, for example Tunisia has two returns of -0.27 and $+0.27$ on 15–16 Sep 2004. Training was done on data from 2004 to 2009 (about 1,350–1,510 returns), validation on 2010–2011 (about 500 returns) and test 2012–2014 (about 750 returns). The stocks have about 2,700 more days in 2015–2025.

The code is written in R and `nnet` is used for MLP. Jobs run in parallel on 16 cores. The code and supplementary materials are available on Github.

Data checks. Following Ismail et al. (2016) and Rubio et al. (2023) we first tested the returns (Table 1). Every series has kurtosis above 3 and the Jarque–Bera test rejects normality with $p < 10^{-4}$. Tunisia's kurtosis of 522 is mostly from the two September 2004 days. The ADF test rejects a unit root for all series and KPSS does not reject stationarity at 5% (Tunisia is borderline with 0.463), so returns can be modelled without differencing. After fitting, the ARCH-LM test on the plain ARMA residuals was very significant ($p \approx 10^{-45}$ for South Africa) while on the standardized GARCH residuals it was not significant, so GARCH effectively captures the clustering.

Table 1. Full-sample daily log returns (2004 to 2014 for indices, to 2025 for stocks). Kurt is raw kurtosis (normal = 3). JB: Jarque–Bera p. ADF and KPSS are statistics; all ADF $p < 10^{-4}$.

Series	n	Skew	Kurt	JB p	ADF	KPSS
S. Africa	2750	-0.18	6.90	$< 10^{-4}$	-25.9	0.147
Tunisia	2720	-0.38	522.53	$< 10^{-4}$	-64.3	0.463
Nigeria	2722	0.00	5.24	$< 10^{-4}$	-25.4	0.180
Infosys	5444	-0.58	14.89	$< 10^{-4}$	-45.2	0.055
IBM	5534	-0.34	12.36	$< 10^{-4}$	-22.9	0.124
Google	5376	0.32	10.67	$< 10^{-4}$	-28.4	0.108

Table 2. Test RMSE $\times 10^4$, 2012–2014, Gaussian GARCH. Matched dates on which all three models have a forecast. Bold is the lowest in each row. Gain is MSE reduction vs. raw (%).

Series	n	Raw	DWT	MODWT	Gain %	
					DWT	MODWT
S. Africa	749	1.3426	1.3448	**1.3394**	-0.3	$+0.5$
Tunisia	745	0.7076	**0.6790**	0.7061	$+7.9$	$+0.4$
Nigeria	745	1.8920	**1.8015**	1.9173	$+9.3$	-2.7
Infosys	736	25.3177	**24.9339**	25.2165	$+3.0$	$+0.8$
IBM	751	**4.1300**	4.1585	4.1305	-1.4	0.0
Google	752	7.7822	7.7857	**7.7781**	-0.1	$+0.1$

5.2. Core result: raw vs. DWT vs. MODWT

Table 2 shows that differences in RMSE are very small. MODWT-GARCH is within about 3% MSE of raw GARCH on every series; it is better on four datasets and worse on Nigeria. Hence, with Haar, the MODWT detail part is a fixed linear filter of the returns, and an ARMA on a filtered series can give same results as what an ARMA on the returns can already do. The paired DWT performs better: it is the best model on Tunisia, Nigeria and Infosys, with 7.9%, 9.3% and 3.0% lower MSE, and close to baseline on the other three stocks. The gain intervals are positive for Nigeria (0.5 to 17.6%) and Infosys (1.4 to 14.9%) but very wide for Tunisia (-20.7 to 14.9%), so the Tunisia gain is not reliable. A possible reason is that the DWT models two coefficient series with their own GARCH, so on the less volatile African markets it can react to short shocks differently. However, on IBM and Google, with high volatility,raw GARCH performs the best

The baseline models used for reference give mixed results(each on its own available dates). It ($\hat{y}_{t+1} = y_t$) is worse than raw GARCH on all six series, for example RMSE 1.951 vs. 1.343 on South Africa. A plain ARMA with constant variance is worse on four series (1.705 on South Africa) but slightly better on Tunisia and Infosys, and the 20-day trailing mean of y even beats GARCH on Tunisia (0.663

Table 3. Raw GARCH test RMSE $\times 10^4$ (2012–2014) under three innovation distributions. Last row indicates the number of the 27 held-out groups (series $\times$ pipeline $\times$ period, incl. 2015–2025) where each distribution has the lowest MSE.

Series	Gaussian	Student-t	GED
S. Africa	1.3426	**1.3421**	1.3423
Tunisia	0.7076	**0.6998**	0.7016
Nigeria	1.8920	**1.8853**	1.8901
Infosys	25.3009	**24.9601**	25.1028
IBM	4.1300	4.1352	**4.1296**
Google	7.7822	7.7715	**7.7649**
Lowest MSE	4	13	10

vs. 0.708). The Tunisia test target is very small and rapidly fluctuating, so a smooth average does well there. Overall the GARCH part does most of the work, and the wavelet step improves it by a small amount at best.Figure 2 shows that the forecasts follow the big spikes of the target only after they happen, which is what we expect from any one-step GARCH-type model.

5.3. Fat-tailed innovations (E1)

Table 3 shows that a fat-tailed distribution improves the results in all of our datasets. Over the 27 held-out groups (6 series $\times$ 3 pipelines in 2012–2014 and 3 stocks $\times$ 3 pipelines in 2015–2025), Student-t has the lowest MSE in 13 and GED in 10, while Gaussian has the lowest in 4 cases. On MAE GED is best in 18 groups. The gains are small, the median MSE change of Student-t vs. Gaussian is about -0.3% in 2012–2014 and -1.2% in 2015–2025, but they mostly go in the right direction. The biggest gain is on Tunisia, which has the fattest tails: DWT-GARCH goes from 0.6790 to 0.6564 with Student-t (6.6% lower MSE). The reason is that with fat tails, a Gaussian likelihood over-reacts to one huge return, and the t or GED likelihood down-weights it, giving more stable α and β.

5.4. Component models, ensembles and MLP (E3–E5)

Figure 3 gives the gains of the extensions on the test. The validation-selected MODWT model (Gaussian, structural or Student-t, as chosen per series) gains between -2.9% and $+1.9\%$, so the E2 order rule did not change much. The selected component model is good on Nigeria ($+2.2\%$) and Infosys ($+2.6\%$) but bad on Tunisia (-22.7%), where the one huge spike in early 2013 hurts a model that works on the target directly. Most of the 24 component variants were much worse than raw GARCH: out of 144 series–variant pairs only 9 beat it and 123 were more than 10% worse. For example on South Africa the db4 and db6 all-ARMA versions had more than 100% higher MSE. Smoothing the target with a two-sided filter and then forecasting at the edge does not work well.

Table 4. MSE gain vs. raw GARCH (%), frozen models. Test = 2012–2014, Later = 2015–2025. Pairwise matched dates.

	Infosys		IBM		Google	
Model	Test	Later	Test	Later	Test	Later
DWT	3.0	2.6	-1.4	1.7	-0.1	-5.5
MODWT	0.8	0.3	0.0	0.1	0.1	-0.3
Ens. base	4.1	1.4	-0.1	-0.5	-0.7	1.4
Ens. comb.	0.8	1.4	-0.9	-0.8	-1.7	1.2
Ens. MODWT	-10.9	-6.6	-11.2	-10.3	-12.3	-7.4
MLP	-812	-52.5	-4.4	-51.1	-5.5	-71.2

The ensembles give mixed results. The base ensemble gains 10.0% on Tunisia and 4.1% on Infosys but loses 17.4% on South Africa. The combined ensemble gains on Tunisia, Nigeria and Infosys. The MODWT ensemble is worse on five of six series (about -11% on the three stocks) because it contains component models that are weak.

The MLP is the weakest model. With seed 101 it is worse than raw GARCH on every series, from -4.4% on IBM to -812% on Infosys, where one bad seed gave a few huge forecasts. The three seeds also disagree a lot: on Infosys the RMSE is 33.27, 24.09 and 33.27 for seeds 101, 202 and 303, against 23.17 for raw GARCH on the same 199 dates. With only about 1,000 noisy training points and 7 inputs, the network can not learn anything that GARCH does not already capture, and it is unstable.

5.5. Sensitivity (E6)

Changing the window(40 or 60 days), using BIC, or raising the order cap to 5 moved the test MSE of the core pipelines by less than about 2.6% in all cases. The window length changed the results a lot. A longer window of 1,536 lowered the MSE of raw GARCH by 7.9% on Tunisia and 7.2% on Nigeria, while a shorter window of 768 increased it by 14.4% on Tunisia. For the stocks the window effect was below 2%. So for these markets, more history gives more stable GARCH parameters, and the choice of window is more important than the choice of wavelet.

5.6. Later stock period, 2015–2025 (E7)

Table 4 runs the same models on eleven more years of stock data. The results are the same. MODWT stays within 0.3% of raw GARCH. DWT keeps its gain on Infosys (2.6%), turns slightly positive on IBM, and is worse on Google. The MODWT ensemble and the MLP perform clearly worse. Small gains can flip sign between periods (for example the ensembles on Google).

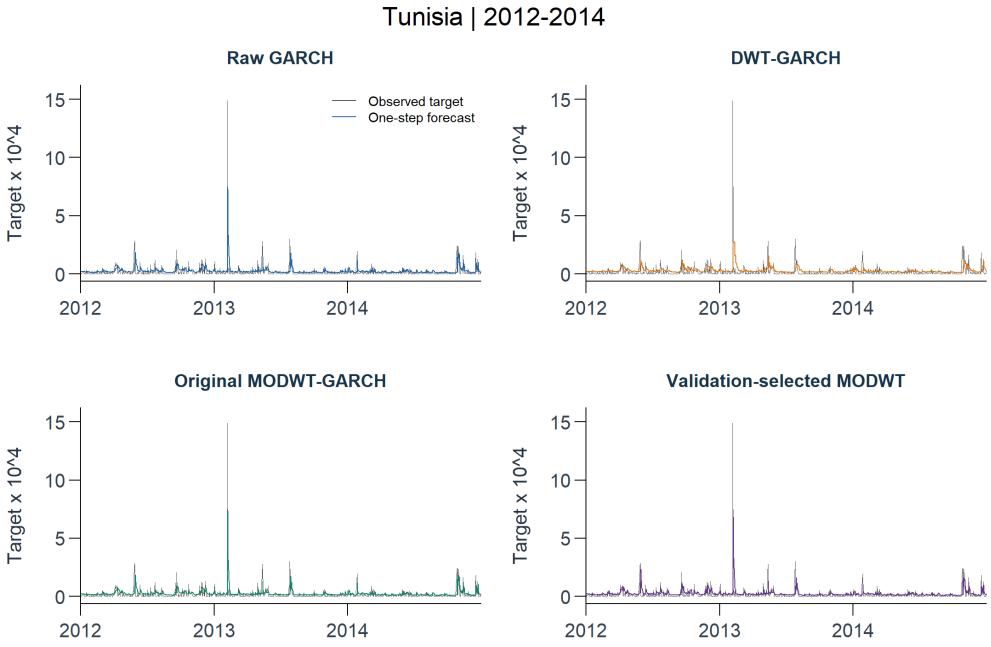


Figure 2. Tunisia, 2012–2014 test: observed target (grey) and one-step forecasts. The "validation-selected MODWT" panel is the Student-t MODWT model chosen by E2/E1 on validation. All models react after the shock in early 2013, not before.

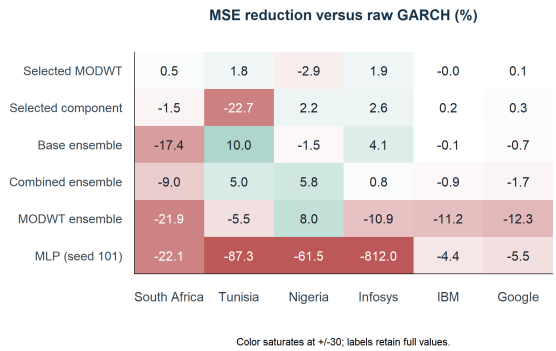


Figure 3. MSE gain vs. raw GARCH (%) on the 2012–2014 test, pairwise matched dates. Green is better, red is worse.

6. Conclusion

We rebuilt the wavelet + ARMA–GARCH approach of Ismail et al. (2016) from scratch and tested it on six markets. Our observations are the following:

- The wavelet step does not improve predictions a lot. MODWT-GARCH performs almost the same as plain GARCH everywhere. DWT-GARCH helps on Nigeria and Infosys, and Tunisia (3–9% lower MSE), but not on IBM, Google or South Africa. So we could not confirm the paper's result that the wavelet model is better on every market.
- How the wavelet forecast is mapped back to the return variance matters a lot. For the paired DWT, conditioning on the already-observed first return of the pair is necessary; without it the variance forecast is far too high.
- Returns do not follow a normal distribution, and Student-t or GED additions give small but consistent gains, the most improvement is seen on the fattest-tailed series (Tunisia).
- A plain GARCH(1,1) is very hard to beat. Ensembles help only when all members perform better, and a small MLP performs worse and is unstable.
- The rolling window length had a bigger effect than the wavelet choice on some markets.

Limitations. The difference in errors of all methods is very small. The obtained target is a noisy proxy for true volatility. We used only Haar wavelets for the core pipelines and first-level transforms. The 2015–2025 stock period is retrospective.

7. Future Work

Our next steps are:

- **Formal tests:** Model Confidence Set,Diebold-Mariano
- **Using Asymmetric Models:** Frameworks like EGARCH or GJR-GARCH can be used to account for the leverage effect, where bad news increases volatility more than good news.
- **Deeper Window:**We can experiment with deeper wavelet levels and other filters inside the return pipelines and weights can be learned on validation instead of equal weights.

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